2026 Grand Union DTP Workshop: Social Network Analysis

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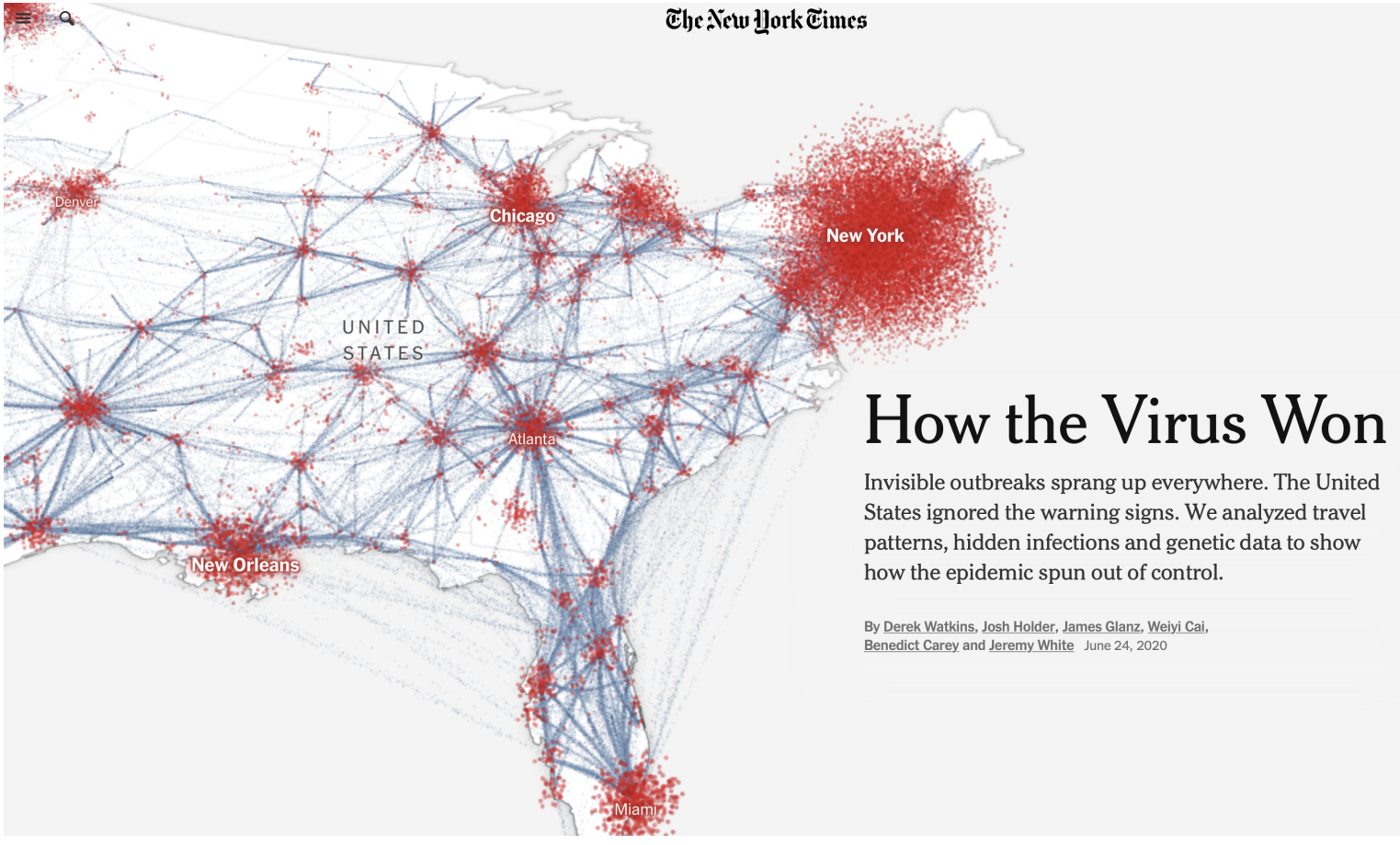


Oxford, 18 March 2026

Outline

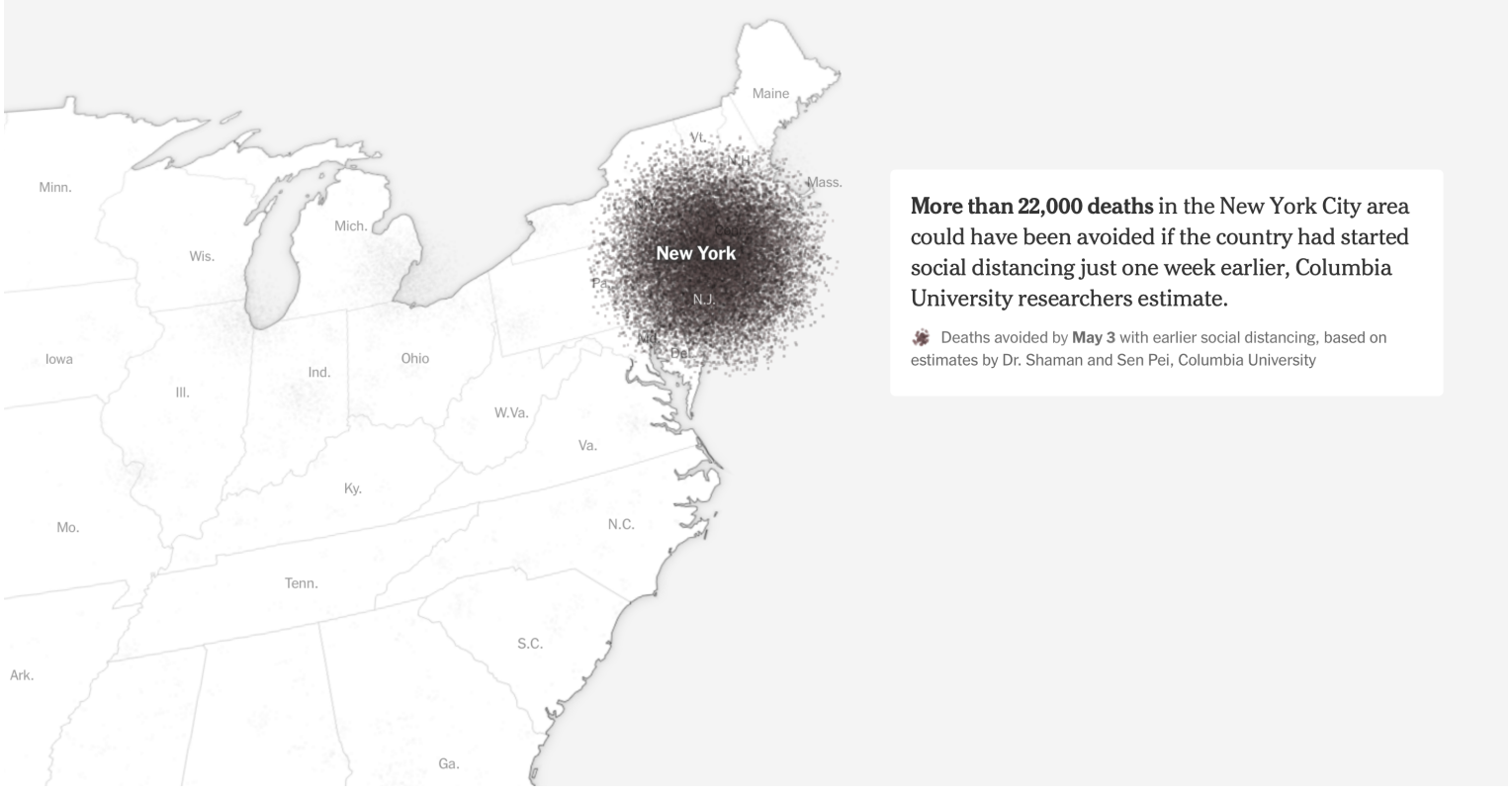
- Getting Started
  - Workshop Introduction
  - Administration
  - Core Readings
- What are Networks?
  - Network Fundamentals
  - A Bit of Graph Theory
  - Key Network Terms
  - Types of Networks
- Why Networks?
  - Theoretical Motivation
  - Network Dynamics
- Representing Network Data
  - Data Formats
  - Network Boundaries
- Discussion & Questions

How the Virus Won

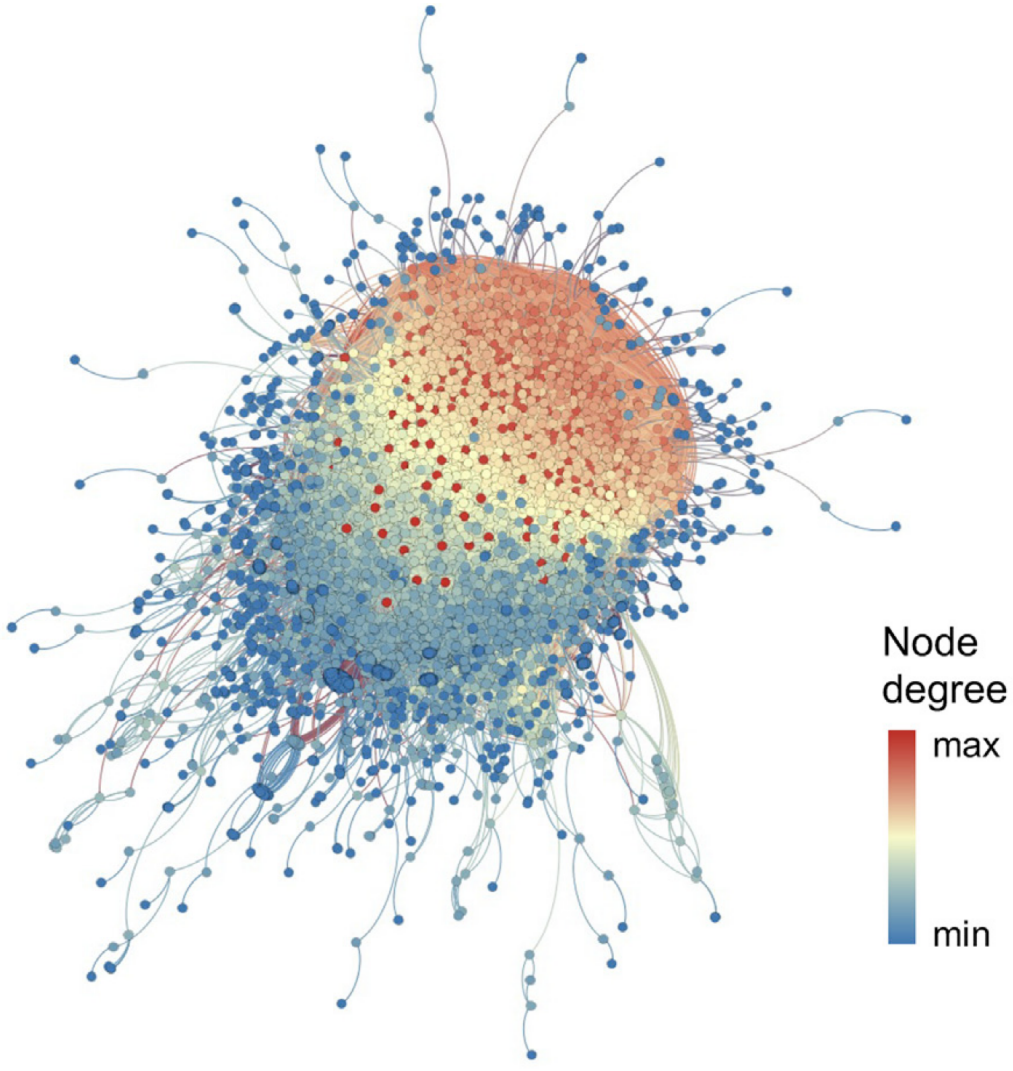


How the Virus Won

The New York Times. 2020. 'How the Virus Won'. The New York Times. https://www.nytimes.com/interactive/2020/us/coronavirus-spread.html



Johnson et al. (2021) Protein–Protein Interaction

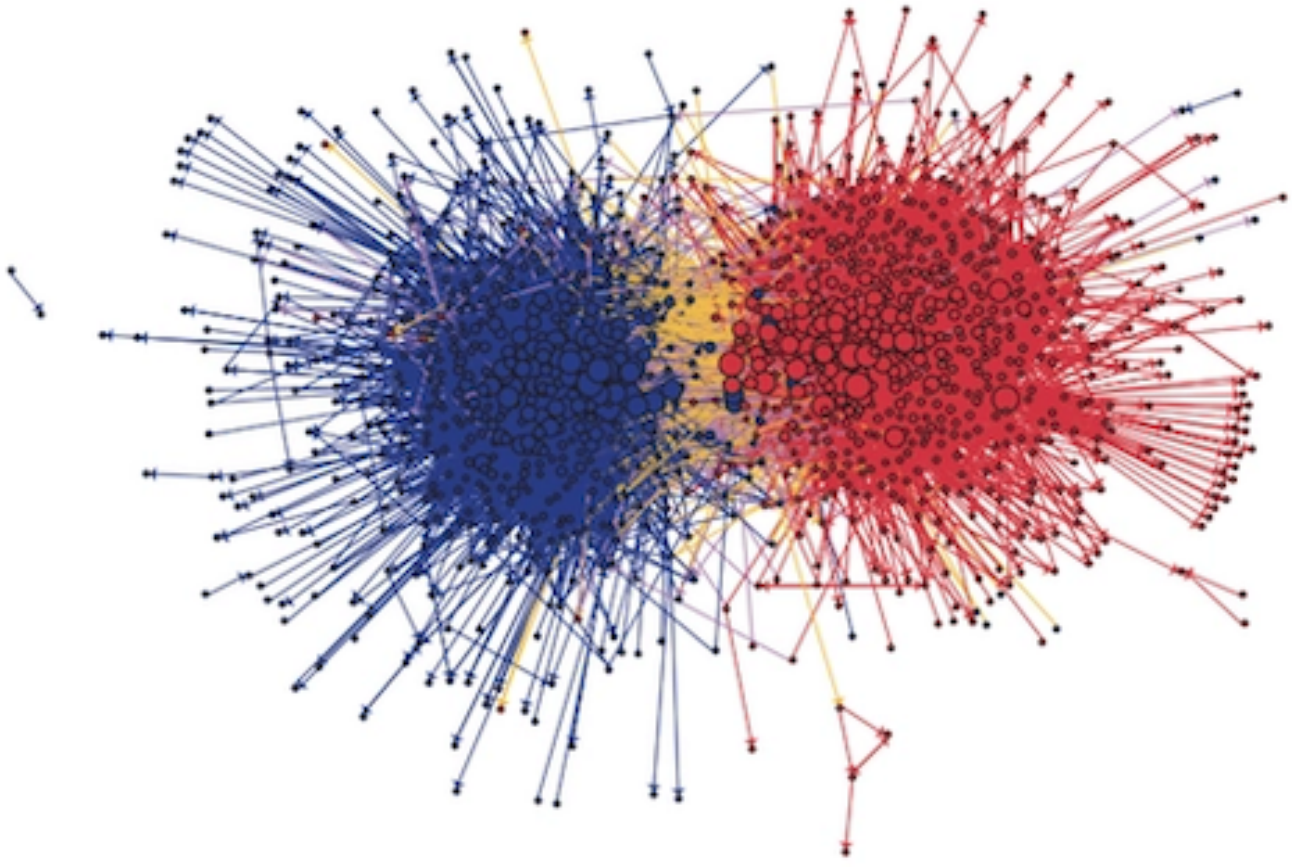


Johnson et al. (2021): Entire PROPER v.1.0 network with proteins as nodes and PPIs as edges. The degree of nodes is colour coded from high (red) to low (blue).

Through these connections, proteins assemble into complexes and pathways that perform biological tasks: gene regulation, metabolism, DNA repair, and signalling.

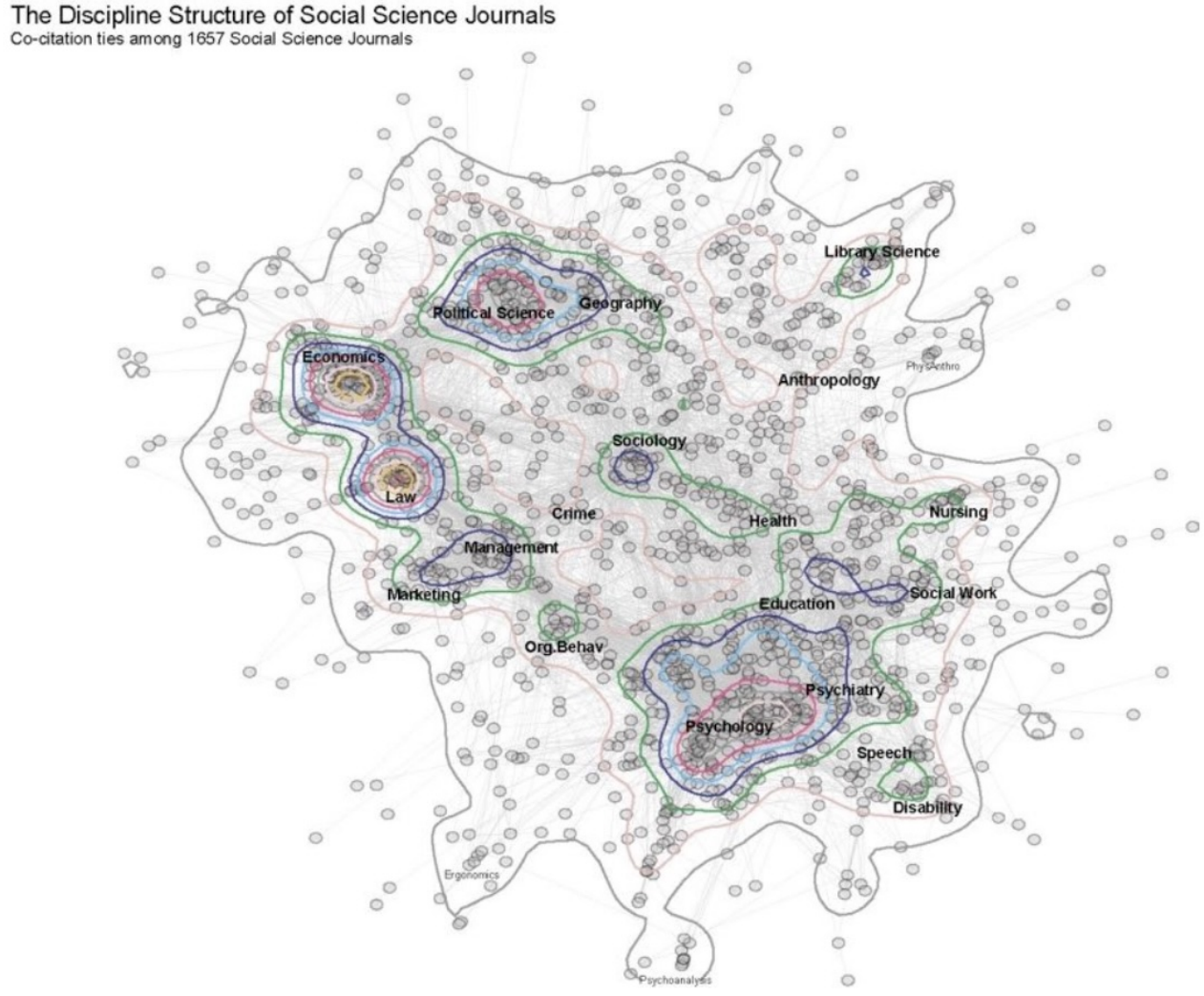
Adamic and Glance (2005)

Visualisation of Democrat and Republican blogs during the 2004 US election.



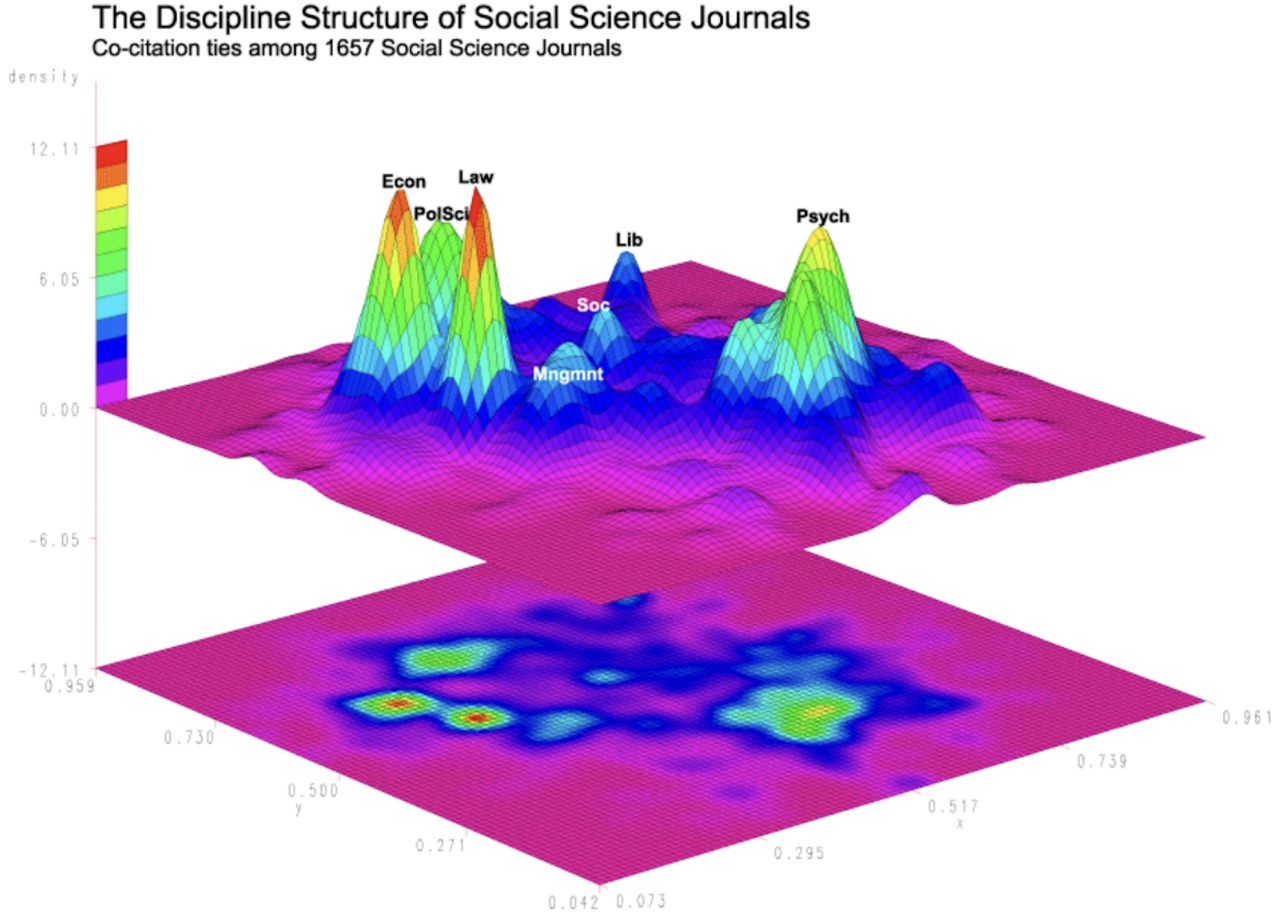
Moody's (2004)

Structure of a Social Science Collaboration Network



Moody's (2004)

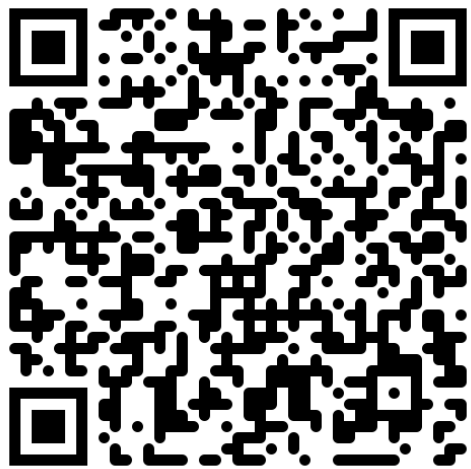
Structure of a Social Science Collaboration Network



WORKSHOP ADMIN

Workshop Materials

Scan to access the full workshop website and resources



https://clemensjarnach.github.io/social_network_analysis_workshop/

Workshop Goals

By the end of this two-day workshop, you will be able to:

- Understand fundamental concepts in network analysis (nodes, edges, graphs)
- Recognise when network methods are appropriate for your research
- Represent network data in multiple formats
- Calculate and interpret network measures
- Detect communities and clusters in networks
- Understand network dynamics and processes
- Apply network analysis to your own research questions

Workshop Outline

Day 1

1. Getting Started
2. What are Networks?
3. Why Networks Matter
4. Representing Network Data

Day 2

1. Network Measures and Metrics
2. Community Detection
3. Network Models and Inference
4. Advanced Topics and Applications

Attendance and Logistics

- Workshop runs 10:00–16:00 on both days
- Lunch breaks at designated times
- All materials available on course page
- Interactive code walkthroughs
- Please bring your laptop with R and RStudio installed

Required Software and Packages

Please ensure you have installed R, RStudio, and the following packages:

```
install.packages(c(
  "tidyverse", "janitor", "readxl", "haven",
  "lubridate", "forcats",
  "igraph", "tidygraph", "ggraph", "statnet",
  "intergraph",
  "RColorBrewer", "viridis"
))
```

Test your installation by loading each package:

```
library(igraph)
library(tidygraph)
library(ggraph)
```

Core Readings

- Avrachenkov and Dreveton (2022) – Statistical foundations
- Burt (2004) – Structural holes and information flow
- Borgatti et al. (2022) – Practical analysis in R
- Granovetter (1973) – Strength of weak ties
- McPherson et al. (2001) – Homophily and assortative mixing

Additional readings will be distributed throughout the workshop.

Github Repository for this Workshop

In this repository, you can find the code, datasets, and all other files used throughout this workshop.

Access Materials Here:

https://github.com/clemensjarnach/social_network_analysis_workshop

WHAT ARE NETWORKS?

Core Idea

A way of thinking about the world that shifts attention from **individual entities** to the **relationships between them**.

Shifting the Question

Classic question:
What is something?

Network question:
How is it connected—and what do those connections imply?

Nodes and Ties

Nodes can be almost anything:

- People, organisations, countries
- Genes, web pages, neurons
- Cities, firms, teams

Ties are the relationships:

- Friendship, collaboration
- Communication, trade
- Trust, conflict, flows

Why this matters

Ties interlink through shared nodes, creating **chains and paths** that connect distant parts of a system—enabling *indirect* influence invisible to attribute-level analysis.

Social Networks Specifically

When nodes are **active social agents** (individuals, teams, firms, cities) connected by **social relationships**, we enter the domain of *social network analysis*.

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Social ties include:
friendship · collaboration · communication · trade · trust · conflict

Levels of Analysis

Dyad
Pairwise relationships between two actors.
Do actors who share business ties also develop affective ones?

Node
Individual positions within the network.
Do people with more connections fare better in job markets?

Network
Global structural properties.
Do tightly connected networks diffuse information faster?

Structure is Causally Consequential

Central theoretical claim

An actor's **position** in a network shapes the constraints and opportunities they encounter. Structure is not merely descriptive—it is *causally consequential*.

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An actor's **position** in a network shapes the constraints and opportunities they encounter. Structure is not merely descriptive—it is *causally consequential*.

The same logic applies at the collective level:

- A sports team of talented individuals may **underperform** if collaboration structures are poor
- An organisation may **fail to innovate** not because members lack creativity, but because information cannot flow across structural gaps between teams

GRAPH THEORY

Networks and Graphs

A **network** and a **graph** are the same thing: a mathematical structure consisting of entities and their relationships.

Formally: $G = (V, E)$

- V = set of vertices (nodes)
- E = set of edges (ties, links)

Networks allow us to model pairwise relations between objects using the language and tools of graph theory, borrowed from mathematics.

Key Terminology

Network / Graph : A set of nodes and the edges connecting them

Node / Vertex : An entity in the network (person, organisation, country, etc.)

Tie / Edge / Arc : The relationship connecting two nodes

Cluster / Community / Group : A subset of nodes more densely connected to each other than to the rest

Key Terminology (continued)

Homogeneous vs Heterogeneous : Whether nodes are of the same or different types

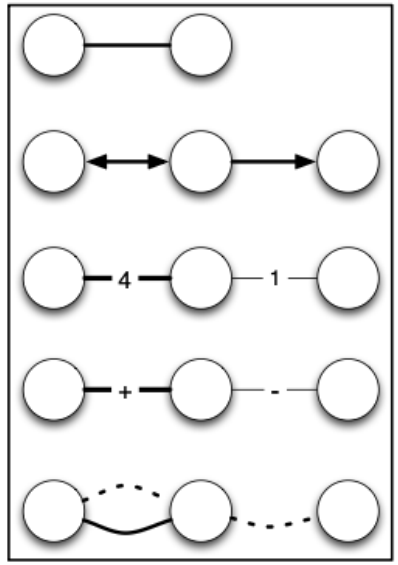
Edge Direction : Whether ties are directed (asymmetric) or undirected (reciprocal)

Edge Weight : Numerical value assigned to an edge (strength, frequency)

Edge Value : Sign of an edge: positive (friend) or negative (foe)

Types of Relationships

Networks can encode different types of ties:



Undirected: "Edge" - e.g., Facebook friends. $V_{ij} = V_{ji}$

Directed: "Arc" - e.g., Gives advice. $V_{ij} \neq V_{ji}$

Weighted: Numerical value of relation, e.g., number of messages sent. Higher value = "stronger" tie between people.

Valued: Positive or negative, e.g., Emotional energy; positive / negative reviews on eBay.

Multiplex: Multiple relation types between people.

Directed ties, reciprocal ties, valued ties, and signed ties each tell a different story.

Defining a Graph:

Consider a graph with 8 vertices:

$V := \{1, 2, 3, 4, 5, 6, 7, 8\}$

And 8 edges connecting them:

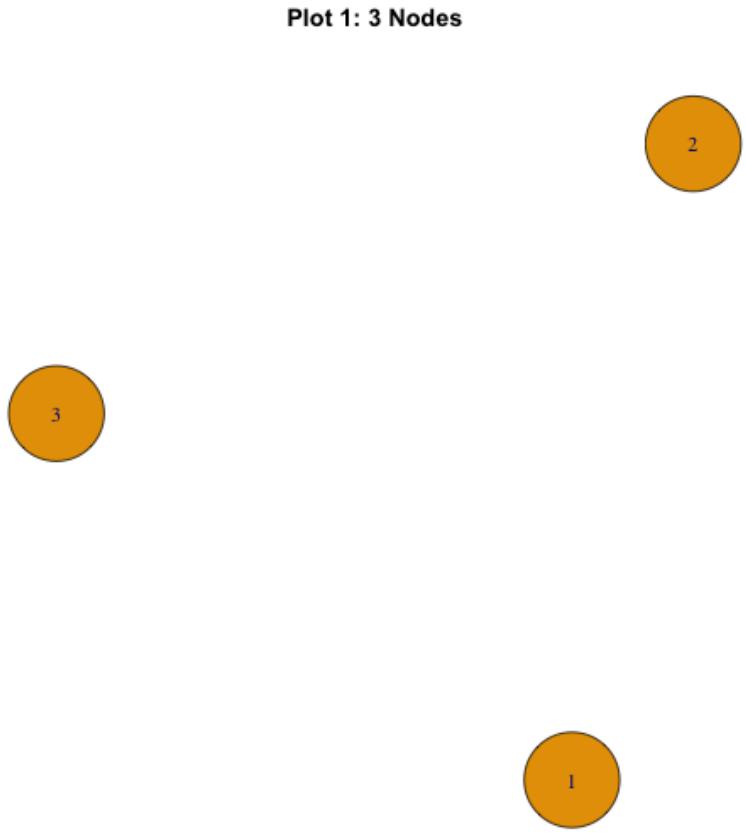
$E := \{\{1, 2\}, \{1, 3\}, \{2, 3\}, \{4, 5\}, \{5, 6\}, \{6, 7\}, \{5, 7\}, \{5, 8\}\}$

We will visualise this network step-by-step using R. Follow along with the interactive code walkthrough:

View R Code

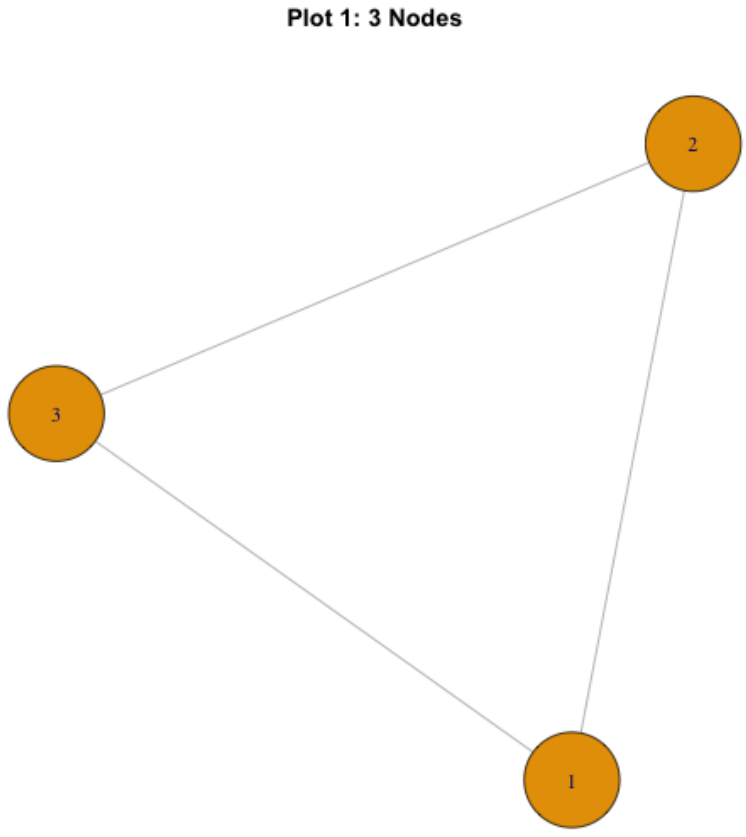
Building a Graph: Three Nodes

Starting simple: three isolated nodes.



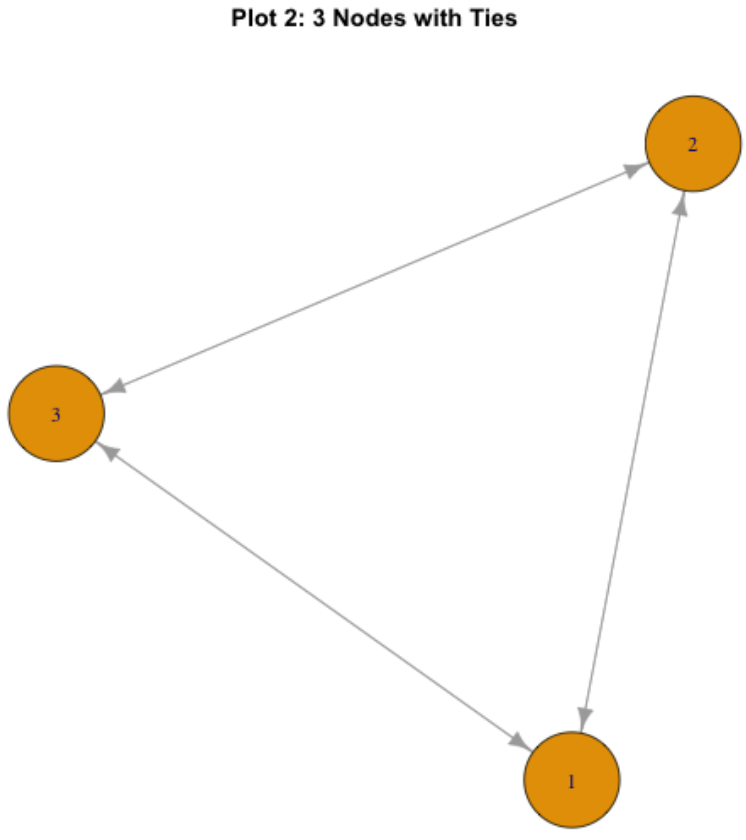
Adding Edges to the Nodes

Now we connect these three nodes with edges.



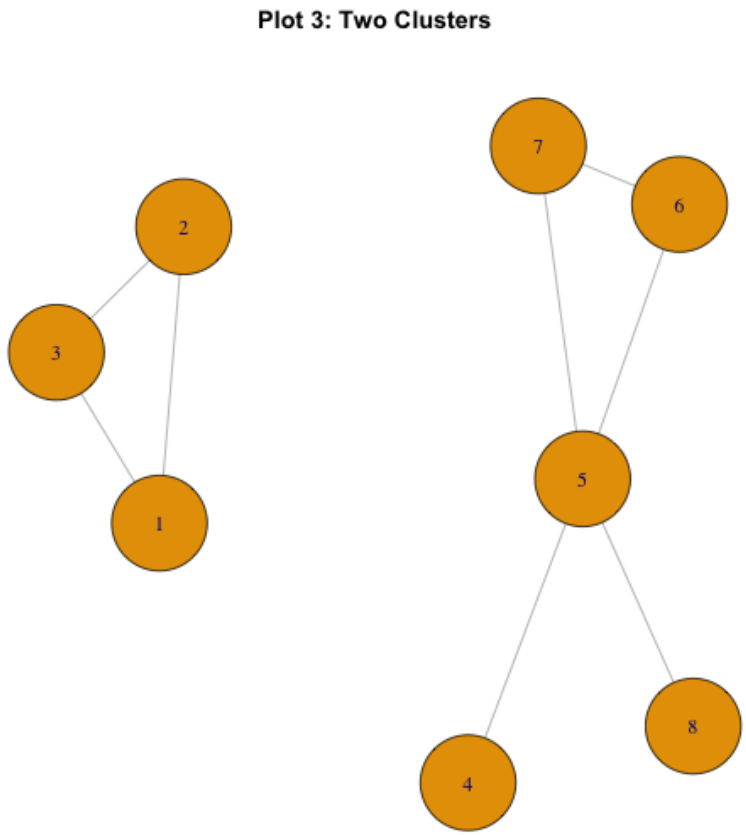
Adding Direction

Each edge represents a relationship between two nodes. We can add arrowheads to show the direction of an edge.



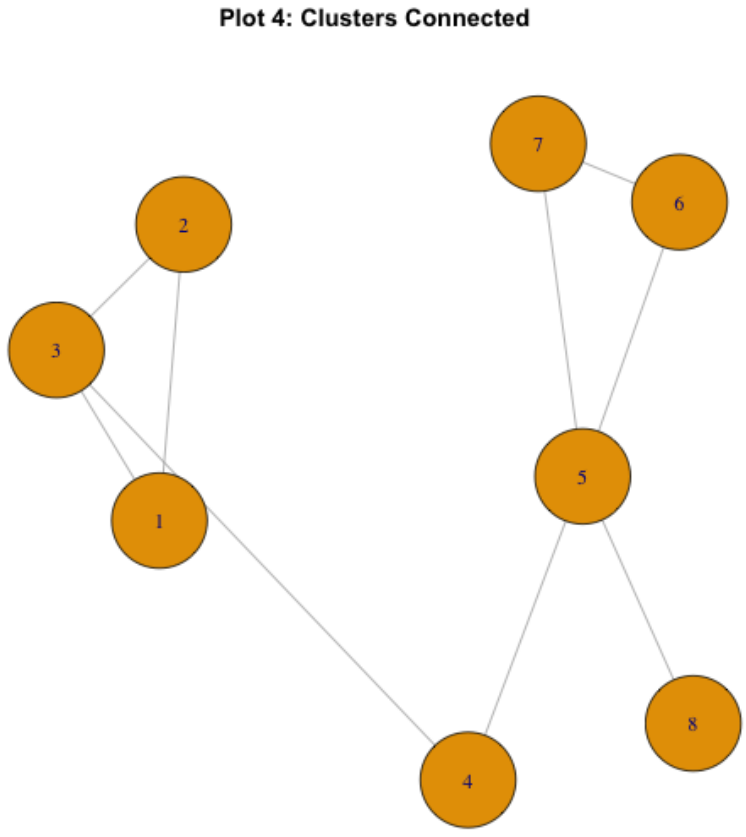
Adding a Second Cluster

We introduce five additional nodes, densely connected among themselves but isolated from the first cluster. Notice the structure: two groups with no bridges between them.



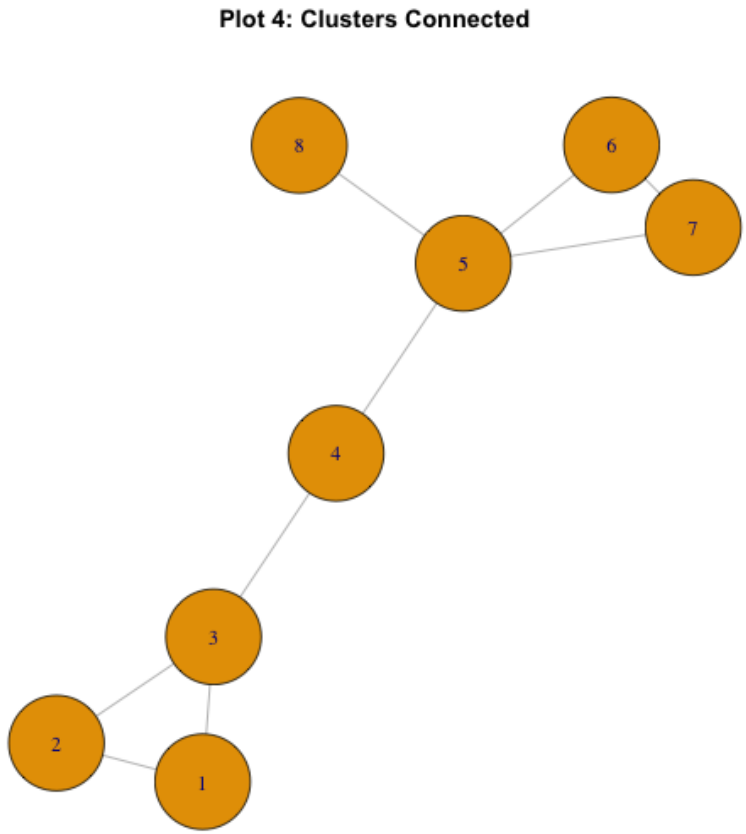
Bridging the Clusters

We add one edge connecting the two clusters.



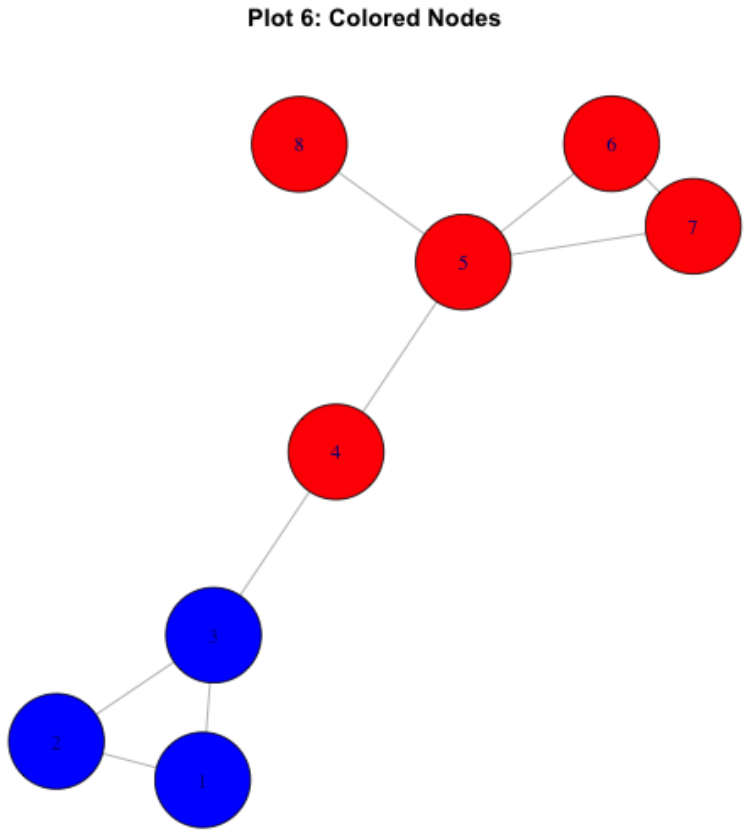
Bridging the Clusters

... This single edge is a **bridge**: critical for flow between groups.



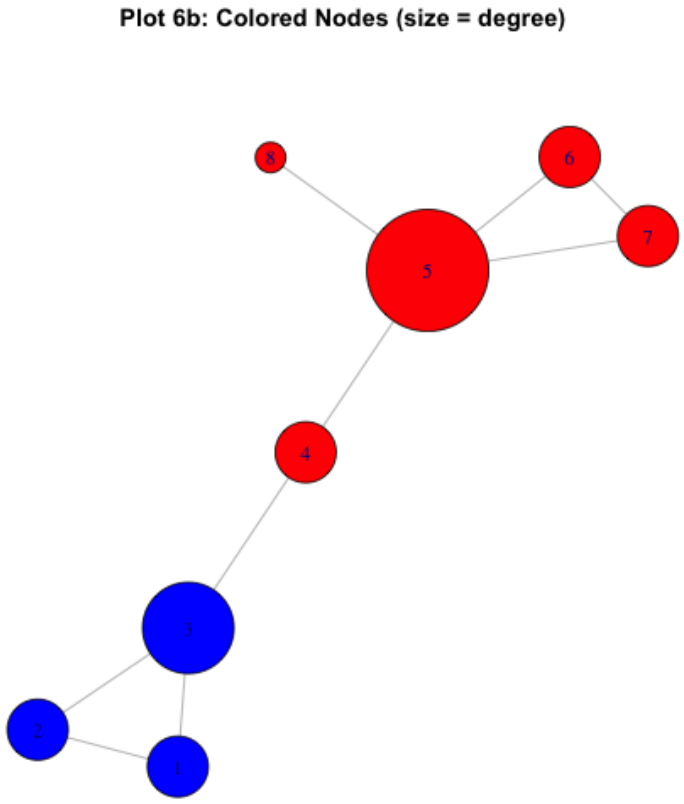
Adding Node Attributes: Colour

We colour nodes by group membership: red for one cluster, blue for the other.



Adding Node Attributes: Size

Node attributes (colour, size, shape) encode additional information beyond network structure.



Building a Graph: Workshop Walkthrough

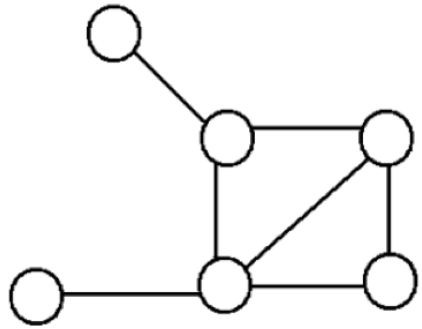
Click to open R Code Walkthrough

Simple Graphs

A **simple graph** has:

- No self-loops (a node cannot be tied to itself)
- No multiple edges (at most one edge between any two nodes)
- Edges are undirected

Most real-world networks are not perfectly simple, but simplicity is often a useful starting assumption.



Directed Graphs

In a **directed graph**, edges have direction.

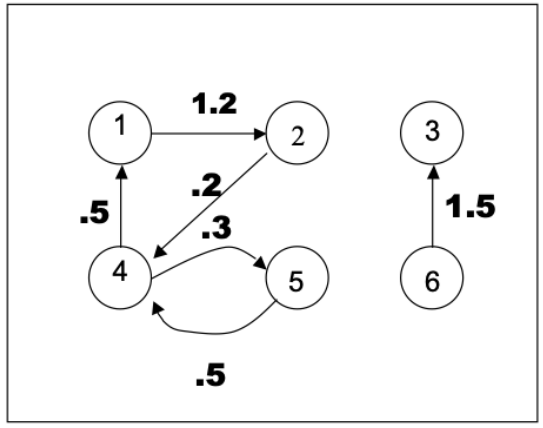
- An edge from i to j (written $i \to j$) is different from $j \to i$
- Examples: follows on Twitter, citations, sending messages, directed trust
- Undirected edges (mutual friendships) can be represented as two directed edges

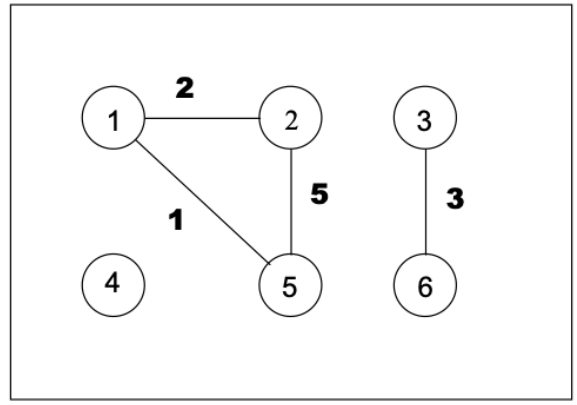
Direction matters for understanding asymmetric relationships.

Weighted Graphs

In a **weighted graph**, edges carry numerical values.

- Weight represents strength, frequency, or intensity
- Examples: number of interactions, duration of contact, confidence in a relationship
- Weights can be continuous or discrete
- Analysis must account for weights; treating all edges equally loses information

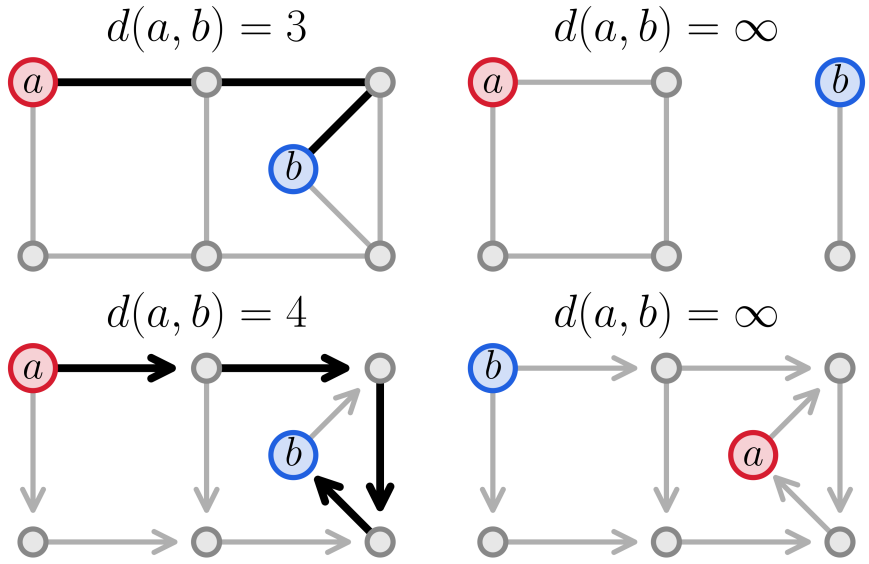




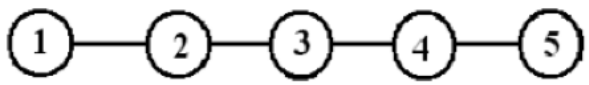
Distance in Networks

The **distance** between two vertices is the number of edges in a **shortest path** (also called a *graph geodesic*) connecting them.

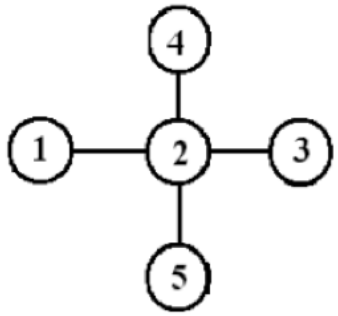
- **Geodesic Distance:** Often used interchangeably with "shortest-path distance."
- **Infinite Distance:** If no path exists between two nodes (e.g., they are in different components), the distance is defined as ∞.
- **Path Length:** Measured by the number of *edges*, not the number of nodes.



Trees and Stars



Paths



Stars

Paths, Trees, and Stars

Path

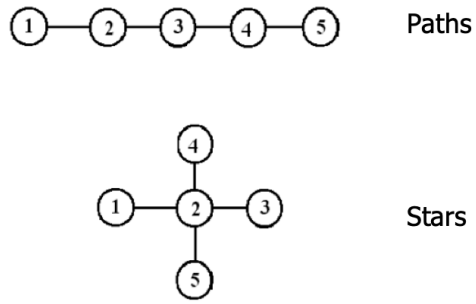
- A path is a sequence of distinct vertices where each adjacent pair is connected by an edge.

Tree Graphs

- A graph in which any two vertices are connected by *exactly one* path.
- **Acyclic:** Contains no cycles or loops.

Star Graphs

- A special tree where one central node connects to all others.
- Represents extreme **centralization**.
- Maximum distance (diameter) is 2.

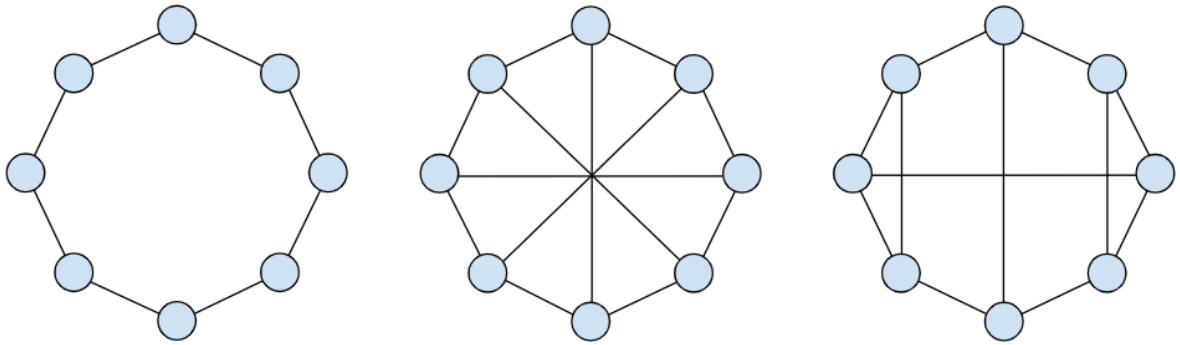


Regular Graph

A **regular graph** is a graph where every vertex has the same number of neighbours. In other words, every vertex has an identical degree k.

Common examples of k-regular graphs include:

- 0-**regular:** An empty graph (no edges).
- 1-**regular:** A collection of disconnected edges.
- 2-**regular:** A collection of disjoint cycles.
- 3-**regular:** Also known as a *cubic graph*.



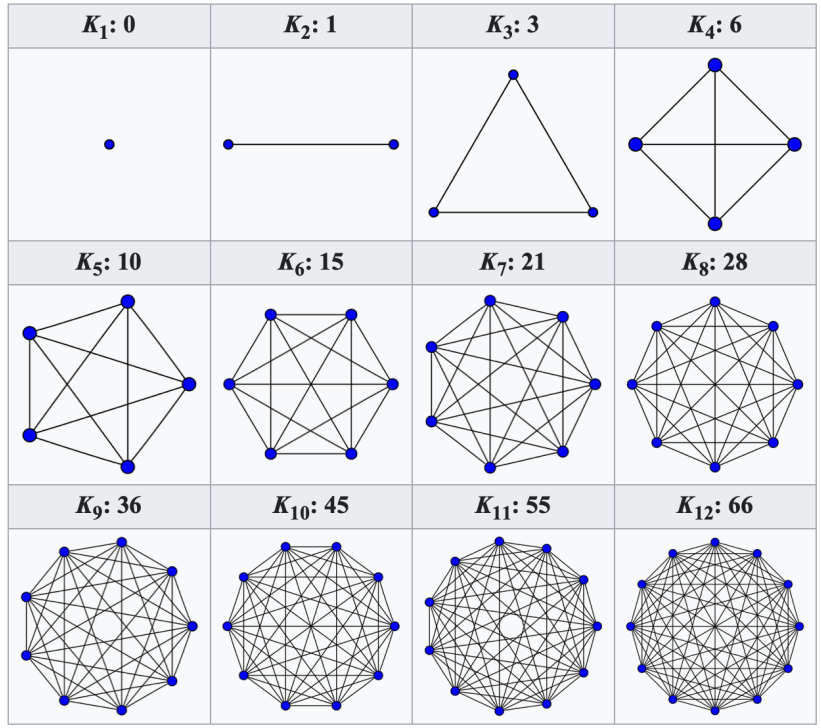
2-regular 3-regular 3-regular

Complete Graph

A **complete graph**, denoted as K_n, is a graph where every pair of distinct vertices is connected by a unique edge. **Key**

Properties of K_n:

- **Regularity:** It is a regular graph of degree $n - 1$.
- **Edge Count:** It contains a total of $\frac{n(n-1)}{2}$ edges.
- **Triangular Numbers:** The number of edges follows the sequence of triangular numbers (e.g., $K_3 = 3, K_4 = 6, K_5 = 10$).



Bipartite (Two-Mode) Networks

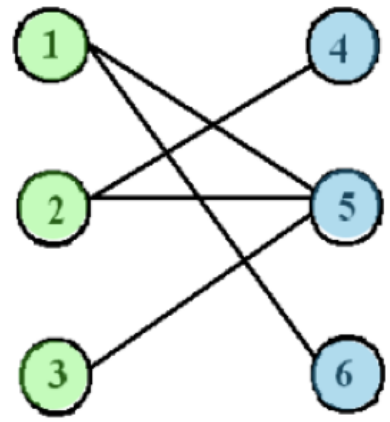
A **bipartite** or **two-mode** network has two distinct sets of nodes, with edges only between sets (not within).

- Example: people and events they attend
- Example: authors and papers they write
- Example: legislators and bills they vote on
- Projection into one-mode networks is common but requires care

Two-mode networks reveal affiliation structures and co-participation.

Bipartite (Two-Mode) Networks

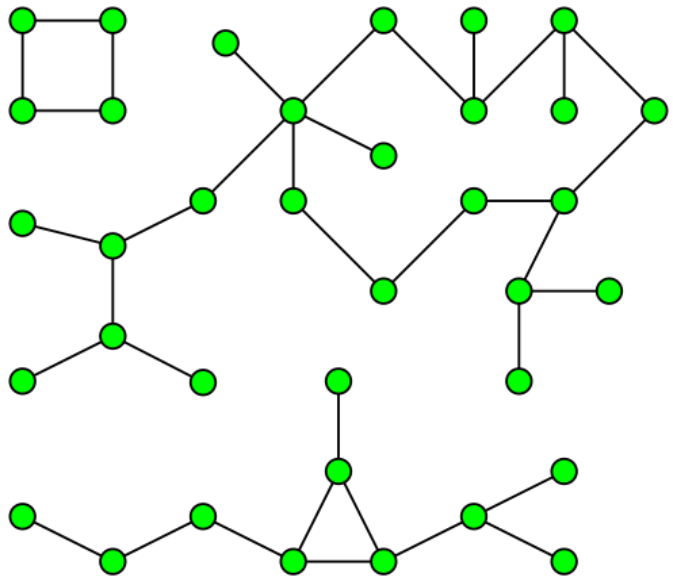
- V can be partitioned into 2 sets V_1 and V_2 such that $(u,v) \in E$ implies
  - either $u \in V_1$ and $v \in V_2$
  - OR $v \in V_1$ and $u \in V_2$.



Component

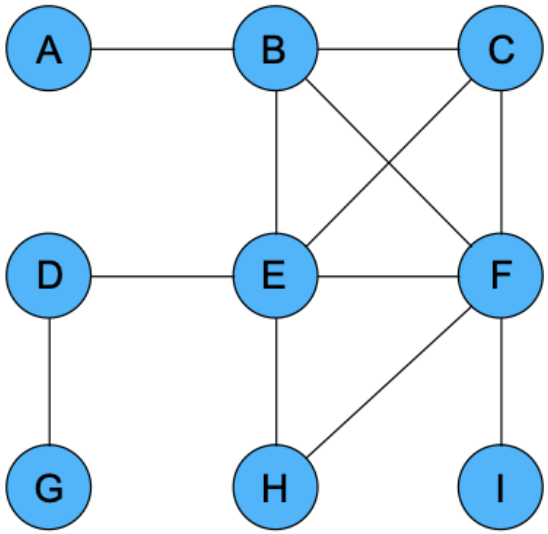
A **component** of an undirected graph is a maximal connected subgraph.

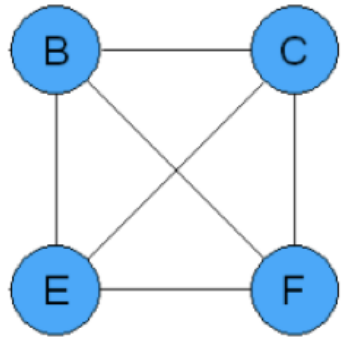
- **Connectivity:** Every node in a component can reach every other node via a path.
- **Maximality:** It is not part of any larger connected group.
- **Partitioning:** Components divide the graph into *disjoint sets* of vertices (no node belongs to two components).



A special subgraph: Cliques

A clique is a subset of vertices in an undirected graph where every distinct vertex pair is adjacent, forming a complete subgraph; it is a maximum complete connected subgraph.



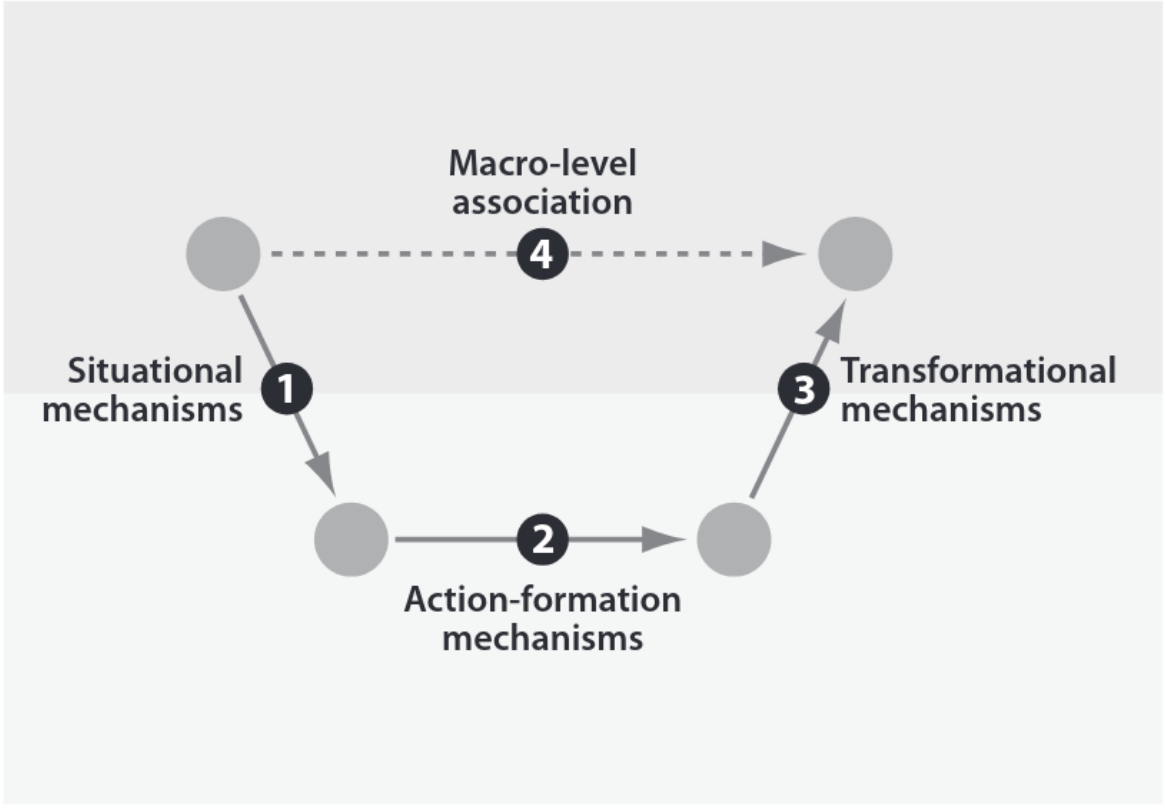


WHY NETWORKS?

Why Network Analysis Matters

1. **Micro to Macro**: Link individual behaviours to emergent collective outcomes (Coleman's boat)
2. **Context Matters**: Individuals cannot be understood in isolation; social context shapes behaviour
3. **New Methods Required**: Standard statistical methods assume independence; networks violate this
4. **Emergent Phenomena**: Swarms, epidemics, cascades, and imitation arise from network structure
5. **Method and Theory**: Network analysis is both a methodological toolkit and a theoretical framework

Coleman's Boat: Micro to Macro

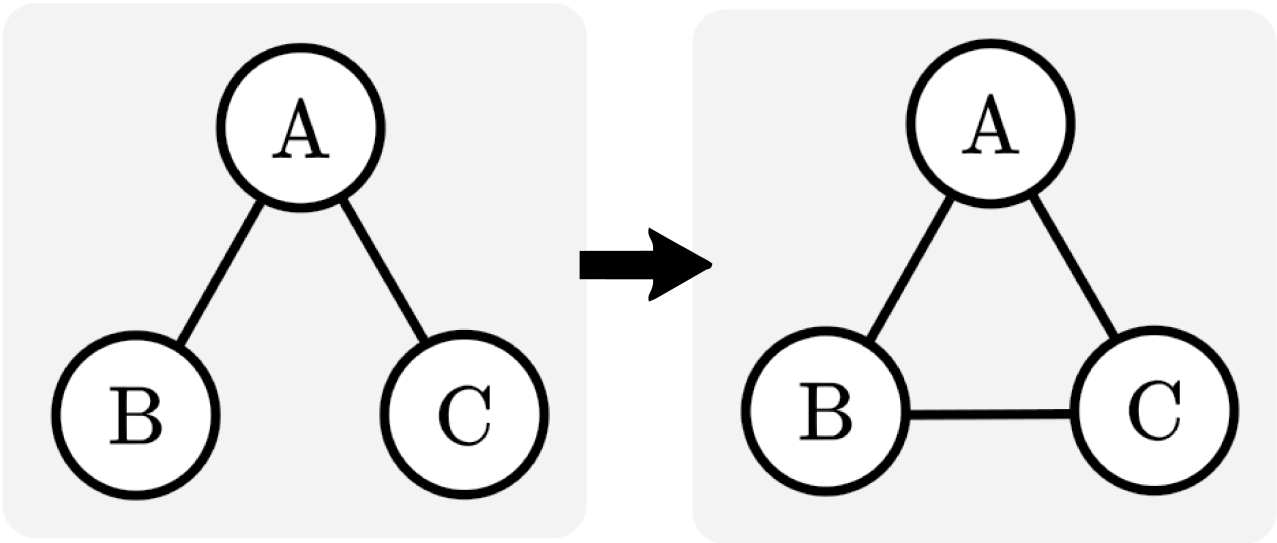


Networks bridge individual actions and system-level outcomes.

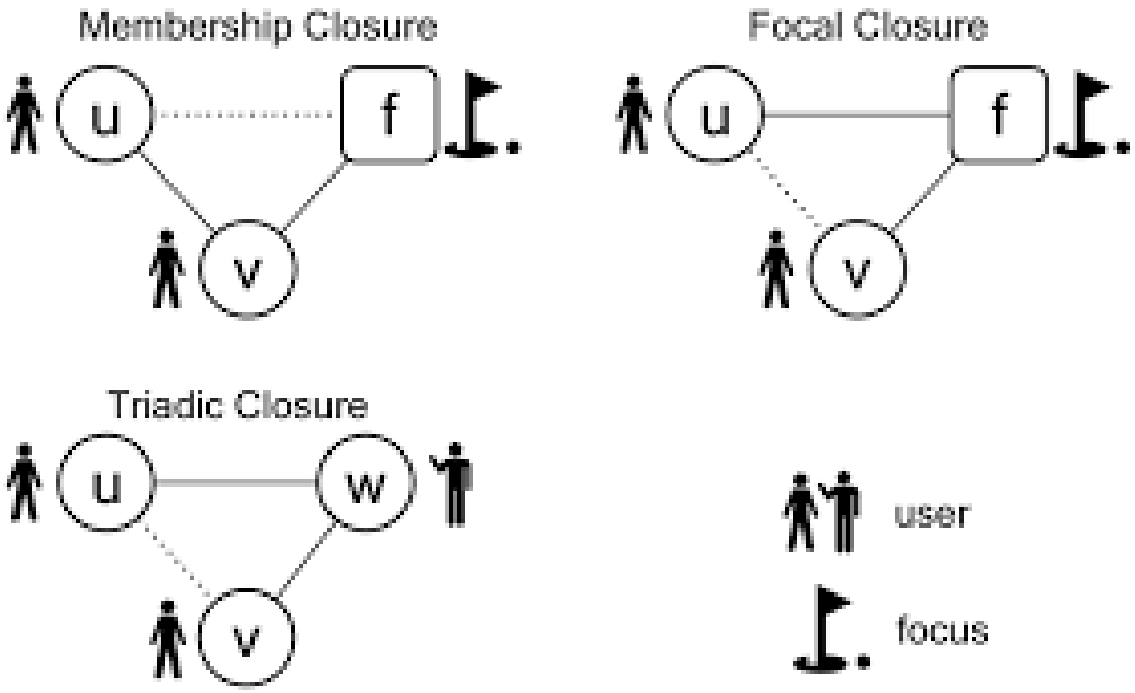
Network Dynamics 1: Triadic Closure

A **triad** is a set of three nodes.

Triadic closure: people tend to want their close friends to know each other. Closing open triads creates clustering (dense local neighbourhoods).



Network Dynamics 1: Triadic Closure II



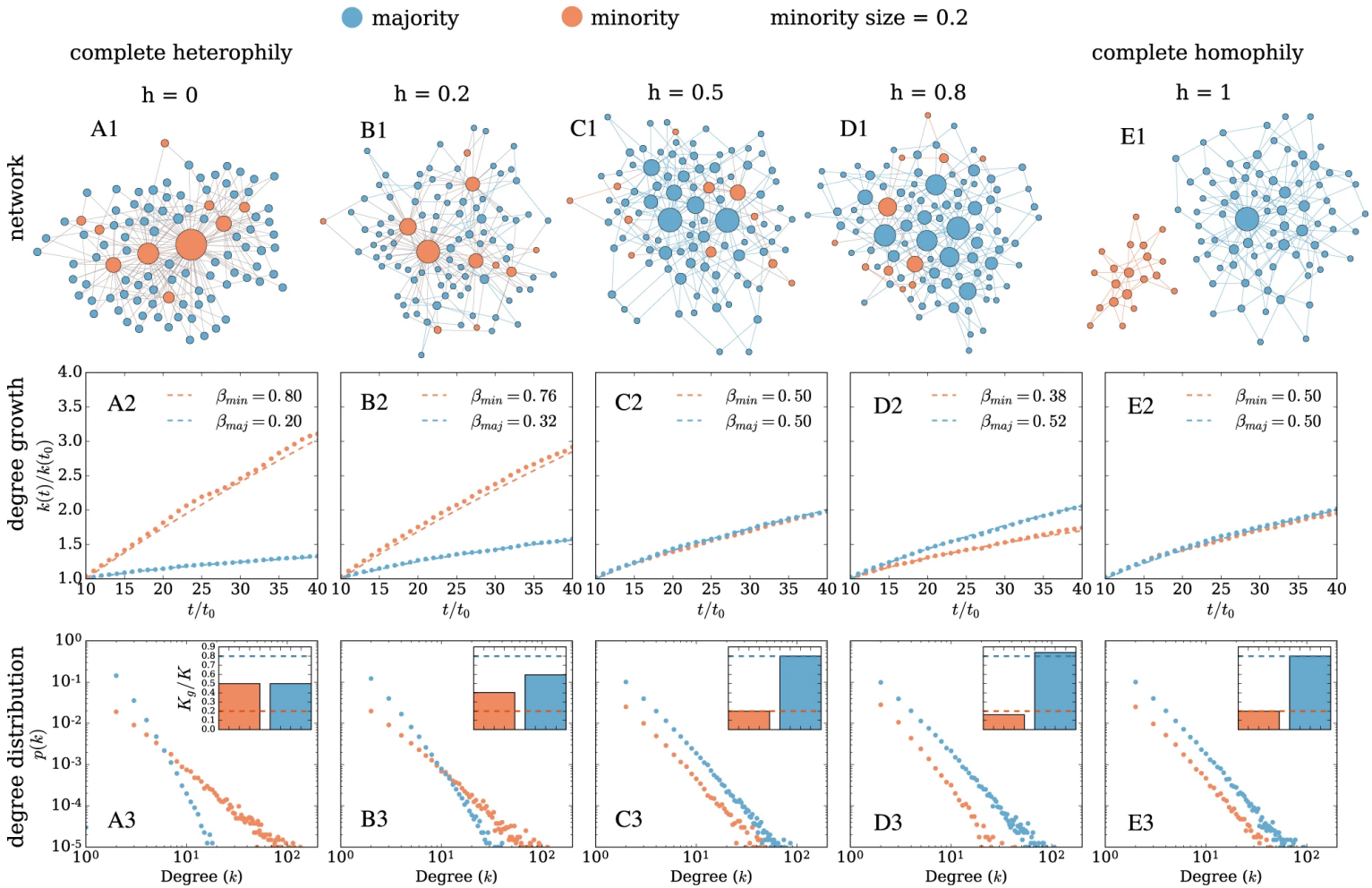
Network Dynamics 2: The Strength of Weak Ties

Granovetter (1973) showed that weak ties are crucial for information flow.

- Strong ties tend to cluster: your close friends know each other
- Information circulates fast within clusters but slowly between them
- Weak ties bridge clusters, providing novel information
- Career advancement, new ideas, and opportunities often arrive via weak ties

This reveals a paradox: the most important connections may be the ones we take for granted as casual.

Network Dynamics 3: Homophily



Source: Karimi, et al., 2018

Empirical Example

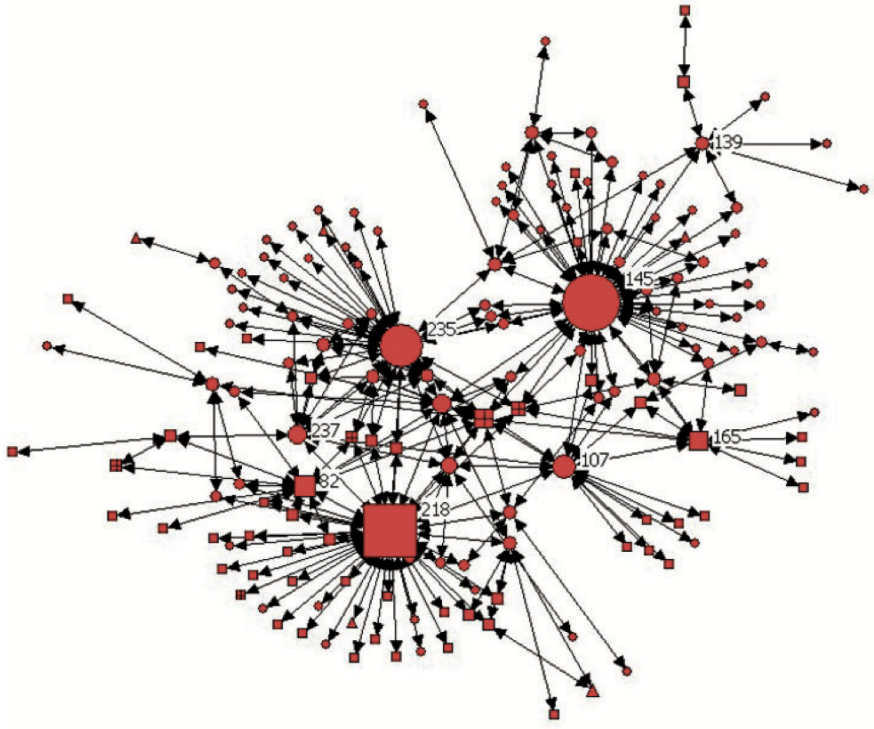


Figure 1 The Russian–Italian Mafia network. Size of node, normalized degree centrality expressed as a percentage; shape, nationality (square, Italian; circle, Russian; triangle, Korean and Polish; box, unknown)

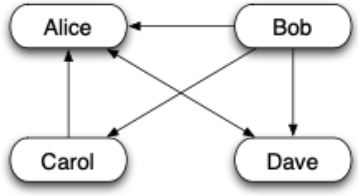
Varese, F. 2013. 'The Structure and the Content of Criminal Connections: The Russian Mafia in Italy'. European Sociological Review 29(5).

REPRESENTING NETWORK DATA

How to Represent Network Data

Networks can be encoded in multiple formats:

	Alice	Bob	Carol	Dave
Alice	-	0	0	1
Bob	1	-	1	1
Carol	1	0	-	0
Dave	1	0	0	-



From	To
Alice	Dave
Bob	Alice
Bob	Carol
Bob	Dave
Carol	Alice
Dave	Alice
Dave	Bob
Dave	Carol

Each format has strengths and weaknesses depending on your data and analysis.

Adjacency Matrices

An **adjacency matrix** is a square matrix where:

- Rows represent "from" nodes, columns represent "to" nodes
- Entry (i, j) indicates whether there is an edge from i to j
- For undirected graphs, the matrix is symmetric
- For weighted graphs, entries contain weights rather than binary indicators

Advantages: easy for matrix algebra, fast for dense networks.
Disadvantages: memory-intensive for large sparse networks, harder to store node attributes.

Edge Lists

An **edge list** is a table with columns for source, target, and optional attributes:

from	to	weight
1	2	0.8
1	5	0.6
2	3	0.9
2	5	0.7

Advantages: compact for sparse networks, easy to add edge attributes.
Disadvantages: requires conversion for some algorithms, less convenient for matrix operations.

Sociograms

A **sociogram** is a visual representation of a network.

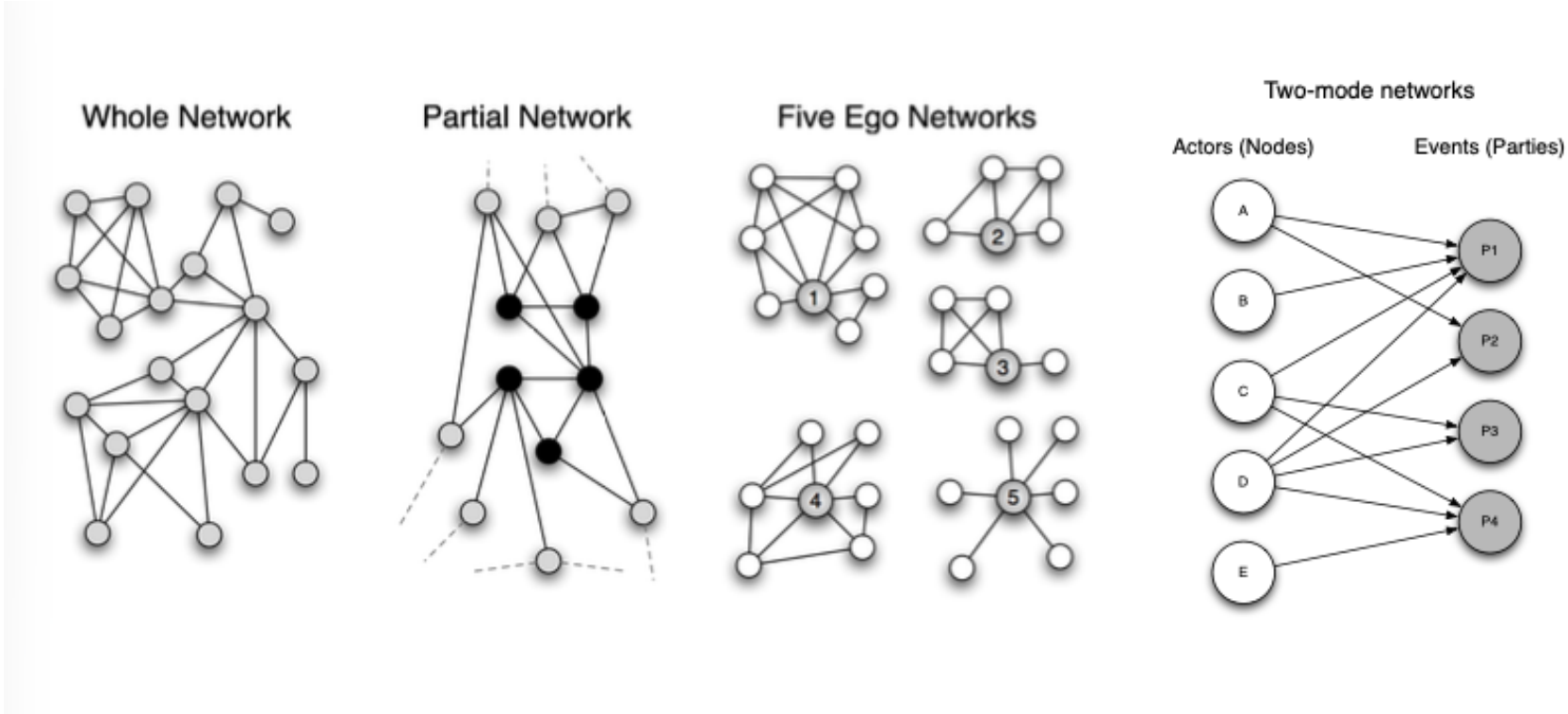
- Nodes drawn as points or circles
- Edges drawn as lines connecting nodes
- Layout algorithm determines node positions
- Visual inspection reveals structure, clusters, bridges

Sociograms are powerful for communication and intuition, but cannot replace quantitative analysis.

Network Collection and Boundaries

Defining the boundary of your network is crucial and often contentious.

- Who or what is included in the set of nodes?
- What types of ties are you measuring?
- Are all ties equally important?



Boundary Decisions: The Hard Problems

- **Inclusion**: Do you include everyone with at least one tie? A threshold? Sampling?
- **Truncation**: Do you observe the full network or only a partial view?
- **Time**: Is the network a snapshot or a dynamic process?
- **Homophily vs heterophily**: Are you studying within-group ties or cross-group ties?

Boundary choices affect inference. Transparency about boundaries is essential.

Random Graphs

Erdős and Renyi (1959)

N nodes

A pair of nodes has probability *p* of being connected.

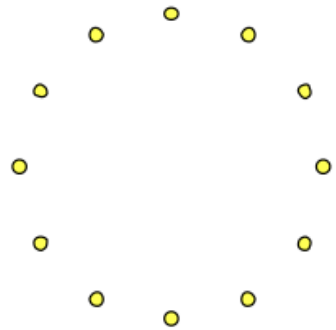
Average degree, $k \approx pN$

What interesting things can be said for different values of p or k ?

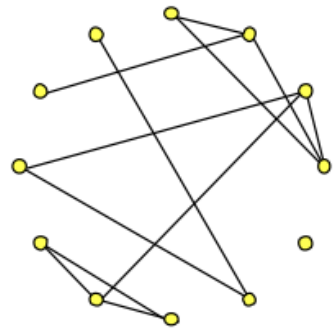
(that are true as $N \to \infty$)

$N = 12$

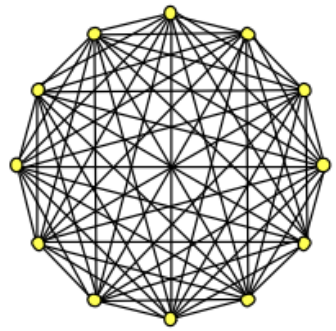
$p = 0.0$; $k = 0$



$p = 0.09$; $k = 1$

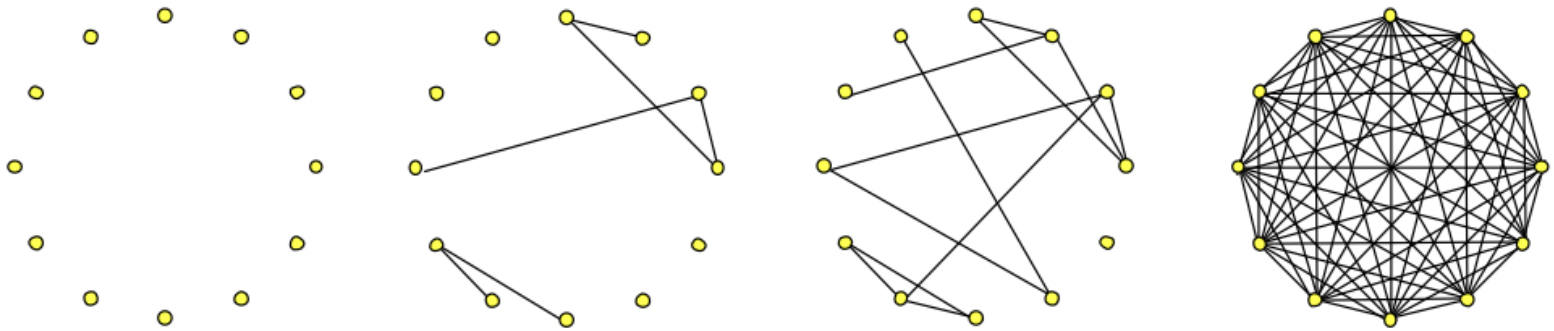


$p = 1.0$; $k \approx \frac{1}{2}N^2$



Random Graphs

Erdős and Renyi (1959)



$p = 0.0$; $k = 0$	$p = 0.045$; $k = 0.5$	$p = 0.09$; $k = 1$	$p = 1.0$; $k \approx \frac{1}{2}N^2$
Size of largest component			
1	5	11	12
Diameter of largest component			
0	4	7	1
Average path length between nodes			
0.0	2.0	4.2	1.0

DISCUSSION & QUESTIONS

Questions

Q & A

Discussion Questions

Structure and mechanism

1. What makes a tie weak? What underlying social mechanisms produce weak versus strong ties?
2. Is there a meaningful distinction between a bridge and a structural hole?

Properties and assumptions

3. Are ties between two individuals always reciprocated?
4. Is information from weak ties always easily accessible?
5. How do you determine the *sign* of a network tie—and are all strong ties strictly positive?

Discussion Questions (continued)

Function and diffusion

6. Is a weak tie always "good"? Consider what is being diffused and to whom.
7. Do online and offline networks differ fundamentally?
8. Have social networks changed over 100 or 1000 years—and if so, why?

Social capital and groups

9. A student argued: "Networks cannot be described as social capital, as social capital can be possessed by an individual whereas networks exist between individuals." Can you apply 'social capital' to a collective—a political party, friendship clique, or Masonic lodge?

Contact

Thank you!

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